

Mohammad Abbas

Full Stack Developer

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 Github

Summary

Full Stack Developer with 2+ years of experience building high-performance full-stack applications using React, Node.js, go, and SQL. Specialized in backend optimization, data-intensive systems, and scalable architectures. Proven ability to design and deploy production systems on GCP using Docker and CI/CD pipelines.

Experience

Full Stack Developer Volunteer, Big Data Specialist	08/2024 – Present Remote
- Designed and deployed LearnDevs, a full-stack platform (Next.js, Go, GCP) aggregating programming courses, remote jobs, tech news and blogs, and GitHub repositories into a unified developer hub. - Built fully automated background job system using cron-based scheduling to ingest tech news, remote jobs, and GitHub repositories daily, eliminating manual content updates. - Built scalable data pipelines using Python and Playwright to aggregate 20,000+ records with scheduled execution and fault-tolerant processing. - Implemented automated testing strategy using Vitest and Playwright, ensuring reliability across frontend and end-to-end workflows. - Implemented CI/CD pipelines with GitHub Actions and containerized deployment workflows for reliable production releases. - Integrated Cloudflare (CDN, DNS, edge security) to optimize global performance and system availability.	
Frontend Developer, Vertex Partners	04/2025 – 10/2025 Remote
- Built and maintained the frontend architecture for places4students.com, a high-traffic student housing platform, using Next.js, improving Lighthouse performance scores to 90+. - Designed scalable UI systems for complex dashboards and map-based interfaces with efficient state management. - Delivered production features in collaboration with backend and product teams in agile cycles.	
Full Stack Developer Intern, 3E Tech	01/2025 – 04/2025 Beirut, Hybrid
- Developed full-stack features for LMS and property management platforms using React, TypeScript, NestJS, PostgreSQL, and Prisma. - Built modular frontend architecture and managed complex client state using Redux Toolkit, improving maintainability of large-scale UI workflows. - Designed and optimized backend services and PostgreSQL queries using Prisma ORM, reducing API response time by ~25%. - Implemented role-based access control and secure validation pipelines across backend services to enforce production-grade authorization rules.	

- Built a high-performance search and filtering system for flight and hotel listings.
- Designed backend authentication and role-based access control system, enforcing secure multi-role workflows across users and admins.
- Contributed to scaling platform features in a collaborative team environment, ensuring system reliability and consistent performance under increasing load.

Skills

Programming Languages: JavaScript, TypeScript, Go, Python, SQL

Frontend: React, Next.js, Tailwind

Backend: Node.js (NestJS, Express), Gin, Flask

Databases: PostgreSQL, MySQL

DevOps: Docker, GCP, GitHub Actions, CI/CD

Automated Testing: Vitest, Playwright

Projects

MoDB

- Built a database engine from scratch in Go, implementing full query lifecycle (lexer → parser → planner → executor).
- Designed custom storage engine with paging and serialization for efficient data persistence.
- Implemented client-server architecture (TCP-based communication) for query execution.
- Demonstrated deep understanding of database internals, system design, and low-level performance tradeoffs.

Custom Programming Language

- Engineered a custom, dynamically-typed programming language from the ground up, implementing parser and grammar using nearley.js to generate an Abstract Syntax Tree (AST).
- Architected a custom transpiler to convert the language's source code into optimized, executable JavaScript, ensuring full compatibility with the Node.js runtime.

Job Match Agent

- Developed a data aggregation system scraping 10+ job platforms in parallel.
- Implemented semantic matching using embeddings + cosine similarity for intelligent job recommendations.
- Designed deduplication and historical tracking systems for data consistency.
- Built backend services in Python with scalable processing pipelines.

Rapid Express

- Engineered a globally-installed CLI tool to automate Express.js app scaffolding, enabling rapid generation of production-ready APIs with a single command.
- Designed a scalable folder architecture with built-in decorators and error-handling middleware, reducing initial developer setup time by 70%.
- Published on npm, achieving 1,200+ downloads and creating a valuable tool for the developer community.

Education

Bachelor in Management Information Systems, *Lebanese University*

2021 – 2025
Lebanon